

Week 3 – Loops Basics

■ Learning Objectives

- Understand repetition and why loops are useful in programming.
- Learn syntax for while loops and for loops.
- Avoid infinite loops by setting proper conditions.
- Use loops to sum, count, and iterate over numbers or lists.

■ Key Concepts

1. **While Loop:** Repeats while a condition is True.
2. **For Loop:** Repeats for a specific number of times using range().
3. **Loop Control:** Use break to exit and continue to skip to next iteration.
4. **Counters and Accumulators:** Track running totals and counts inside loops.

Example 1 – While Loop Counting

```
count = 1
while count <= 5:
    print("Count =", count)
    count += 1

print("Loop finished!")
```

Example 2 – For Loop with range()

```
for i in range(1, 6):
    print("Number:", i)

# range(start, stop) → stops before stop value
```

Example 3 – Summing Numbers 1 to N

```
n = int(input("Enter a number: "))
total = 0

for i in range(1, n + 1):
    total += i

print("Sum =", total)
```

■ Practice Problems

Practice 1 – Countdown Timer

Write a program that counts down from 10 to 1 and then prints 'Blast off!'.

Hint: Use range(10, 0, -1).

Solution:

```
for i in range(10, 0, -1):
    print(i)
print("Blast off!")
```

Practice 2 – Sum of Even Numbers

Compute the sum of all even numbers between 1 and 100.

Solution:

```
total = 0
for i in range(2, 101, 2):
    total += i

print("Sum of even numbers:", total)
```

Practice 3 – Find Factorial

Ask user for n and compute n! using a loop.

Solution:

```
n = int(input("Enter a number: "))
fact = 1

for i in range(1, n + 1):
    fact *= i

print(f"The factorial of {n} is {fact}")
```

■ Loop Control Statements

```
# Break Example
for i in range(1, 10):
    if i == 5:
        break
    print(i)
print("Stopped at 5")
```

```
# Continue Example
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

■ Homework Challenge

Write a program that asks for N and prints a multiplication table (1 to 10) for that number.

Expected Solution:

```
n = int(input("Enter a number: "))

for i in range(1, 11):
    print(f"{n} x {i} = {n * i}")
```